

MI-Lab Ü 2 Walkthrough and further documentation

Preparation: Installation instructions with Docker, see GitHub Readme.

Walkthrough of the transformation of CSV files to FHIR with Mirth-Connect. Each with screenshots and further explanations and test cases.

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1.2) Channel

Create the channels (on the left under "Channel Tasks"). You can choose any name you like.

Patient Channel: MII patient

The screenshot shows the Mirth Connect Administrator interface for editing a channel named 'MII-Patient'. The left sidebar contains navigation links for Dashboard, Channels, Users, Settings, Alerts, Events, and Extensions. The 'Channel Tasks' section includes Save Changes, Export Channel, Debug Channel, and Deploy Channel. The 'Other' section includes Notifications, View User Guide, View User API, View Client API, Help, About Mirth Connect, Visit nextgen.com, Report Issue, and Logout.

The main configuration area is titled 'Edit Channel - MII-Patient' and includes the following sections:

- Channel Properties:** Name: MII-Patient, Enabled: ☒, Id: ce0a04ef-2b83-4513-b9b3-9f4351cad443, Revision: 0, Last Modified: 2024-12-17 09:49:23. Data Types: Set Data Types, Dependencies: Set Dependencies, Initial State: Started, Attachment: None, Store Attachments: ☒, Tags: Enter channel tag.
- Message Storage:** Development, Content: All, Metadata: All, Durable Message Delivery: On, Performance: [Progress Bar], Encrypt message content: ☐, Remove content on completion: ☐ Filtered only, Remove attachments on completion: ☐.
- Message Pruning:** Metadata: Store indefinitely, Prune metadata older than [] days, Content: Prune when message metadata is removed, Prune content older than [] days, Allow message archiving: ☒, Prune Errored Messages: ☐ (incomplete, errored, and queued messages will not be pruned).
- Custom Metadata:** Table with columns: Column Name, Type, Variable Mapping. Rows: SOURCE (STRING, mirth_source), TYPE (STRING, mirth_type). Buttons: Add, Delete, Revert.
- Channel Description:** Text area for channel description.

At the bottom, it shows 'Connected to: https://localhost:8443 as admin | 8:49 AM UTC (UTC +0)'.

Encounter Channel: MII Encounter

The screenshot shows the Mirth Connect Administrator interface for editing a channel named 'MII-Encounter'. The left sidebar contains navigation links for Dashboard, Channels, Users, Settings, Alerts, Events, and Extensions. The 'Channel Tasks' section includes Save Changes, Export Channel, Debug Channel, and Deploy Channel. The 'Other' section includes Notifications, View User Guide, View User API, View Client API, Help, About Mirth Connect, Visit nextgen.com, Report Issue, and Logout.

The main configuration area is titled 'Edit Channel - MII-Encounter' and includes the following sections:

- Channel Properties:** Name: MII-Encounter, Enabled: ☒, Id: 79eac2cd-84c8-4e68-9cd2-c105c6fa4578, Revision: 0, Last Modified: 2024-12-17 09:51:43. Data Types: Set Data Types, Dependencies: Set Dependencies, Initial State: Started, Attachment: None, Store Attachments: ☒, Tags: Enter channel tag.
- Message Storage:** Development, Content: All, Metadata: All, Durable Message Delivery: On, Performance: [Progress Bar], Encrypt message content: ☐, Remove content on completion: ☐ Filtered only, Remove attachments on completion: ☐.
- Message Pruning:** Metadata: Store indefinitely, Prune metadata older than [] days, Content: Prune when message metadata is removed, Prune content older than [] days, Allow message archiving: ☒, Prune Errored Messages: ☐ (incomplete, errored, and queued messages will not be pruned).
- Custom Metadata:** Table with columns: Column Name, Type, Variable Mapping. Rows: SOURCE (STRING, mirth_source), TYPE (STRING, mirth_type). Buttons: Add, Delete, Revert.
- Channel Description:** Text area for channel description.

At the bottom, it shows 'Connected to: https://localhost:8443 as admin | 8:51 AM UTC (UTC +0)'.

Complete channels for import are available on GitHub at: <https://github.com/IMISE/MI-Lab-E02-CSV-to-FHIR/tree/main/MaterialChannels/>

1.3.1) Source Connector

Switch to the "Source" tab in the channel. Configure the source connector as follows.

Explanation: The directory "/opt/connect/appdata/" in the Docker environment is mapped externally to the directory "Setup/mirth-connect/". This allows Mirth Connect to read all data and subfolders located in the "Setup/mirth-connect/" folder.

MII-Patient Source Connector:

Connector Type: File Reader

Polling Settings

Schedule Type: Interval Next poll at: Donnerstag, Jan 9, 11:31:15 AM

Poll Once on Start: ☐ Yes ☒ No

Interval: seconds

Source Settings

Source Queue: OFF (Respond after processing)

Queue Buffer Size:

Response: None

Process Batch: ☒ Yes ☐ No

Batch Response: ☐ First ☒ Last

Max Processing Threads:

File Reader Settings

Method: file Test Read

Advanced Options: <None>

Directory:

ftp:// /

Filename Filter Pattern: ☐ Regular Expression

Include All Subdirectories: ☐ Yes ☒ No

Ignore . files: ☒ Yes ☐ No

Anonymous: ☒ Yes ☐ No

Username:

Password:

Timeout (ms):

Secure Mode: ☒ Yes ☐ No

Passive Mode: ☒ Yes ☐ No

Validate Connection: ☒ Yes ☐ No

After Processing Action: Move

Move-to Directory:

Move-to File Name:

Error Reading Action: None

Error in Response Action: After Processing Action

Error Move-to Directory:

Error Move-to File Name:

Check File Age: ☒ Yes ☐ No

File Age (ms):

File Size (bytes): - ☒ Ignore Maximum

Sort Files By: Date

File Type: ☐ Binary ☒ Text

Encoding: Default

channelName

channelId

DATE

COUNT

UUID

SYTIME

originalFilename

MII-Encounter Source Connector:

Connector Type: File Reader

Polling Settings

Schedule Type: Interval Next poll at: Donnerstag, Jan 9, 11:31:15 AM

Poll Once on Start: ☐ Yes ☒ No

Interval: seconds 

Source Settings

Source Queue: OFF (Respond after processing)

Queue Buffer Size:

Response: None

Process Batch: ☒ Yes ☐ No

Batch Response: ☐ First ☒ Last

Max Processing Threads:

File Reader Settings

Method: file Test Read 

Advanced Options: <None>

Directory:

ftp:// /

Filename Filter Pattern: ☐ Regular Expression

Include All Subdirectories: ☐ Yes ☒ No

Ignore . files: ☒ Yes ☐ No

Anonymous: ☒ Yes ☐ No

Username:

Password:

Timeout (ms):

Secure Mode: ☒ Yes ☐ No

Passive Mode: ☒ Yes ☐ No

Validate Connection: ☒ Yes ☐ No

After Processing Action: Move

Move-to Directory:

Move-to File Name:

Error Reading Action: None

Error in Response Action: After Processing Action

Error Move-to Directory:

Error Move-to File Name:

Check File Age: ☒ Yes ☐ No

File Age (ms):

File Size (bytes): - ☒ Ignore Maximum

Sort Files By: Date

File Type: ☐ Binary ☒ Text

Encoding: Default

channelName
channelId
DATE
COUNT
UUID
SYSTIME
originalFilename

Test: To test whether Mirth-Connect can find the folder, click on "Test Read" under "File Reader Settings" → "Method". If Mirth-Connect can find the folder, the following message will appear: Successfully connected to: /opt/connect/appdata/Patient or Encounter.

1.3.2) Destination Connector

Switch to the "Destinations" tab in the channel:

MII-Patient Destination Connector:

Connector Type: **HTTP Sender** ☐ Wait for previous destination

- Destination Settings

Queue Messages: ☒ Never ☐ On Failure ☐ Always

Advanced Queue Settings: 0 Retries

Validate Response: ☐ Yes ☒ No

Reattach Attachments: ☒ Yes ☐ No

- HTTP Sender Settings

URL: **Test Connection**

Use Proxy Server: ☐ Yes ☒ No

Proxy Address:

Proxy Port:

Method: ☒ POST ☐ GET ☐ PUT ☐ DELETE ☐ PATCH

Multipart: ☐ Yes ☒ No

Send Timeout (ms):

Response Content: ☒ Plain Body ☐ XML Body

Parse Multipart: ☒ Yes ☐ No

Include Metadata: ☐ Yes ☒ No

Binary MIME Types: ☒ Regular Expression

Authentication: ☐ Yes ☒ No

Authentication Type: ☒ Basic ☐ Digest ☐ Preemptive

Username:

Password:

Query Parameters: ☒ Use Table ☐ Use Map:

Name	Value
------	-------

Headers: ☒ Use Table ☐ Use Map:

Name	Value
------	-------

Content Type:

Data Type: ☐ Binary ☒ Text

Charset Encoding: **UTF-8**

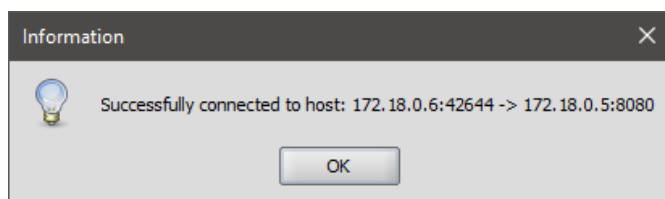
Content:

MII-Encounter Destination Connector:

The changes here are the same as in the MII Patient Destination Connector. Therefore, there are no images here.

Explanation URL: The URL of the Hapi FHIR server is specified here as "<http://hapi:8080/fhir/>". Since Mirth-Connect and the Hapi FHIR server are located in the same Docker Compose network, they can communicate via the container name and the internal ports. If you want to access the FHIR server via the browser/curl, you must use "<http://localhost:8090/fhir/>" because this happens outside the Docker Compose network and therefore the ports that are open to the outside must be used. For further explanations, see the Docker documentation: <https://docs.docker.com/compose/how-tos/networking/>

Test: Click "Test Connection" to test whether Mirth-Connect can connect to the Hapi FHIR server. The following should be returned:

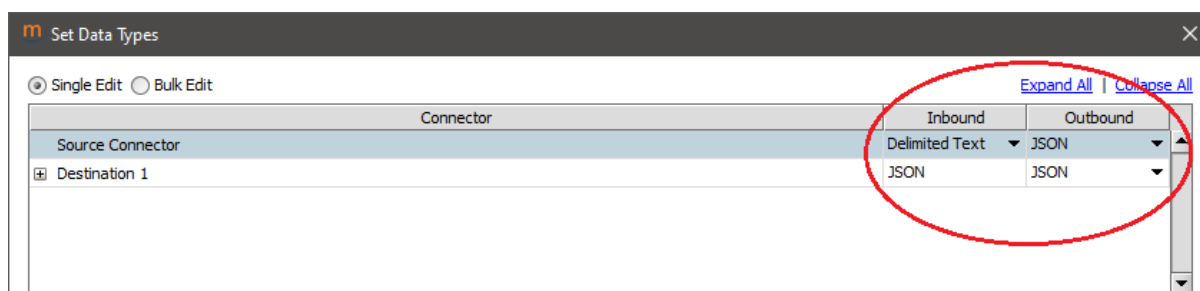


1.3.3) Data Types

Switch to the "Summary" tab in the channel.

Under "Set Data Types," set the inbound and outbound as follows:

MII Patient:



MII Encounter:

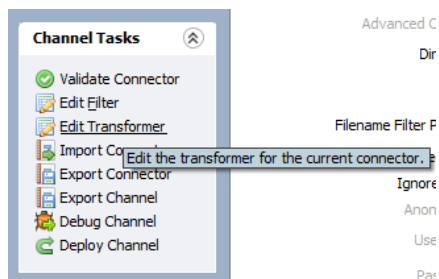
The changes here are the same as for MII Patient. Therefore, there are no images here.

1.4) Importing transformers

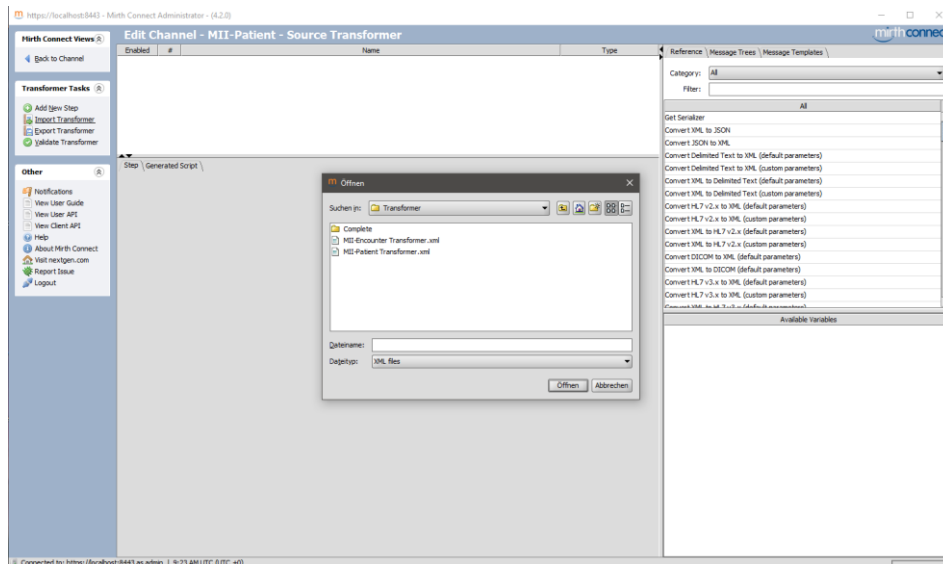
Switch to the "Source" tab in the channel.

Import transformers

Click on "Channel Tasks" > "Edit Transformer."

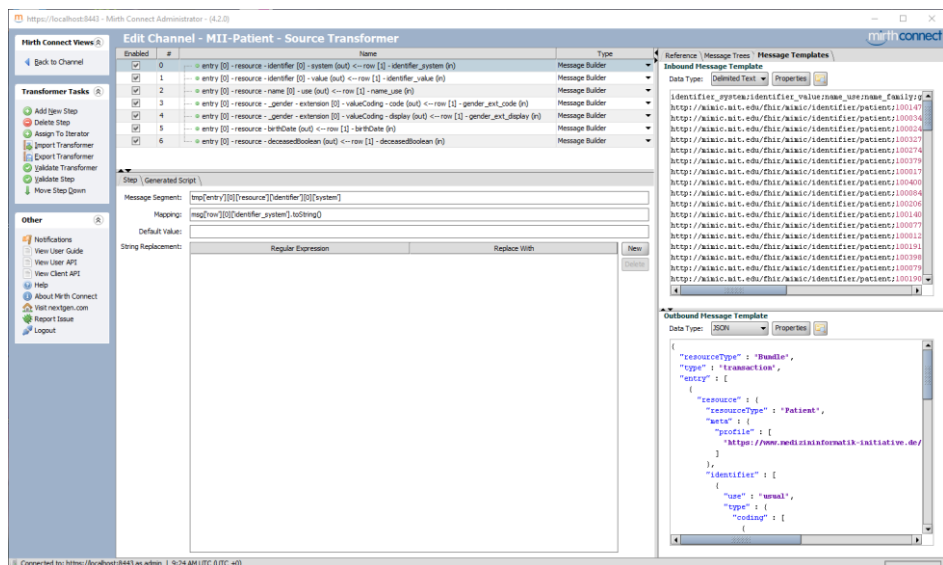


Import the respective transformer under "Transformer Tasks" > "Import Transformer":



MII patient transformer:

Transformer imported successfully. Here, the "Message Template" tab was selected on the right:



MII Encounter Transformer:

Transformer Tasks

- Add New Step
- Delete Step
- Assign To Iterator
- Import Transformer
- Export Transformer
- Validate Transformer
- Update Step
- Move Step Down

Other

- Notifications
- View User Guide
- View User API
- View Client API
- Help
- About Mirth Connect
- Visit nevigen.com
- Report Issue
- Login

Enabled	#	Name	Type
☒	0	entry [0] - resource - class - display (out) <- row [1] - ClassDisplay (n)	Message Builder
☒	1	entry [0] - resource - serviceProvider - identifier - value (out) <- row [1] - ServiceProvider (n)	Message Builder
☒	2	entry [0] - resource - period - start (out) <- row [1] - start (n)	Message Builder
☒	3	entry [0] - resource - period - end (out) <- row [1] - end (n)	Message Builder
☒	4	entry [0] - resource - hospitalization - admission - coding [0] - code (out) <- row [1] - AdmissionSource (n)	Message Builder
☒	5	entry [0] - resource - hospitalization - dischargeDisposition - coding [0] - code (out) <- row [1] - DischargeDisposition	Message Builder

Step 0 | Generated Script

Message Segment: msg[entry[0][resource][class][display]]

Mapping: msg[row[0][ClassDisplay].toString()

Default Value:

String Replacement:

Regular Expression	Replace With

Message Templates

Inbound Message Template

Data Type: Delimited Text

Properties

Message Logs:

- 241811552finished:100042192EMER/emergency/TRANSFER FROH H
- 219261192finished:100096192EMER/emergency/TRANSFER FROH H
- 23983182finished:10018081EMER/emergency/TRANSFER FROH H
- 22942076finished:10004033EMER/emergency/TRANSFER FROH H
- 21696140finished:10031442EMER/emergency/TRANSFER FROH H
- 20626031finished:10005817EMER/emergency/TRANSFER FROH H
- 20297619finished:10019389EMER/emergency/TRANSFER FROH H
- 24982425finished:10002495EMER/emergency/TRANSFER FROH H
- 20755971finished:10038081EMER/emergency/TRANSFER FROH H
- 22265261finished:10019917EMER/emergency/TRANSFER FROH H
- 24216660finished:10037961EMER/emergency/TRANSFER FROH H
- 24547356finished:10026354EMER/emergency/TRANSFER FROH H
- 26352758finished:10019460EMER/emergency/TRANSFER FROH H
- 23931430finished:10020740EMER/emergency/TRANSFER FROH H
- 27089790finished:10012552EMER/emergency/TRANSFER FROH H
- 28335091finished:10014354EMER/emergency/TRANSFER FROH H
- 27487225finished:10014354EMER/emergency/TRANSFER FROH H

Outbound Message Template

Data Type: JSON

Properties

```
{
  "resourceType": "Bundle",
  "type": "transaction",
  "entry": [
    {
      "resource": {
        "resourceType": "Encounter",
        "meta": {
          "profile": [
            "http://www.medrxivinformatics-initiative.org/Encounter"
          ]
        },
        "status": "finished",
        "identifier": [
          {
            "type": "coding": [
              {
                "code": "1000000000"
              }
            ]
          }
        ]
      }
    }
  ]
}
```

Important: If the templates are not displayed on the right as shown in the respective second screenshots, click on the "Message Template" tab in the upper right corner.

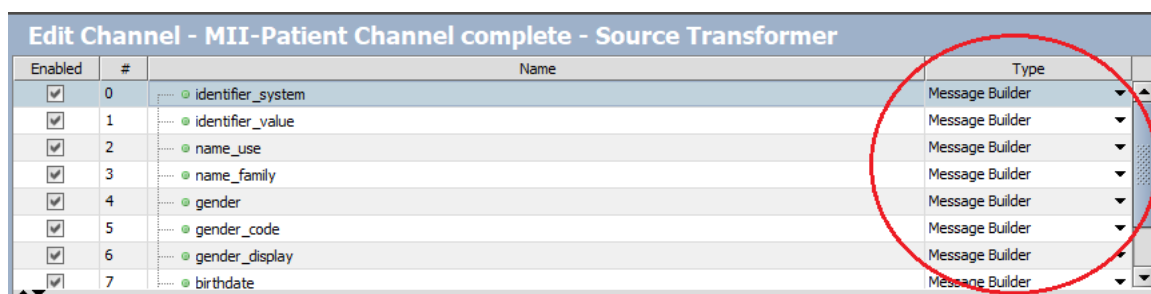
1.5) Configuring the transformers

This is where the actual mapping step for the transformation from CSV to FHIR takes place. The previously imported transformers consist of several steps. Each step is equivalent to an attribute to be mapped. In the task, two mapping steps must be added to the other existing steps. So, two steps for "MII Patient Transformer" and two steps for "MII Encounter Transformer."

Important: After adding a new step, the type to the right of "Mapper" must be changed to "Message Builder"!

Message Builder - Replaces attribute from outbound FHIR template with attribute from inbound CSV data record.

Mapper - Maps attribute from a message to a variable in Mirth-Connect. This is not intended in this exercise.



Enabled	#	Name	Type
☒	0	identifier_system	Message Builder
☒	1	identifier_value	Message Builder
☒	2	name_use	Message Builder
☒	3	name_family	Message Builder
☒	4	gender	Message Builder
☒	5	gender_code	Message Builder
☒	6	gender_display	Message Builder
☒	7	birthdate	Message Builder

1.5.1) MII patient

a) Name_family

Under "Transformer Tasks," click "Add New Step":

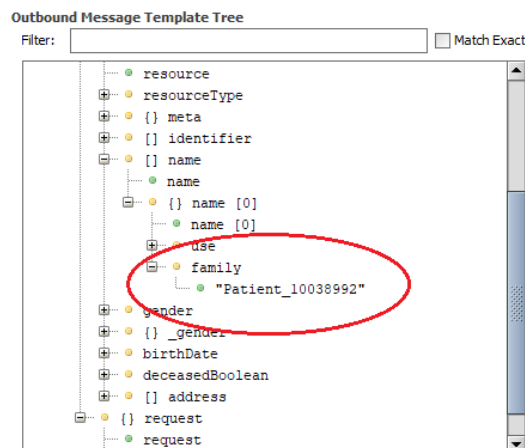
Change the type to "Message Builder"!

Tip: If you select the "Message Trees" tab on the right, you can drag **and** drop the respective segments into the fields. Only fields with a green dot can be selected. Here for "**name_family**":

(top)



(bottom)



Drag the "name_family" field from the Inbound Message Template Tree (left image) in the Transformer Step into the "Mapping" field:

Step \ Generated Script \

Message Segment:

Mapping:

Default Value:

String Replacement:

Regular Expression	Replace With

New Delete

Now drag the field "Patient_100389992" under name->family in the Outbound Message Template (image on the right) in the Transformer Step into the "Message Segment" field:

Step \ Generated Script \

Message Segment:

Mapping:

Default Value:

String Replacement:

Regular Expression	Replace With

New Delete

The finished Transformer step for the attribute **name_family** of the patient now looks like this:

Edit Channel - MII-Patient Channel complete - Source Transformer

Enabled	#	Name	Type
☒	2	name_use	Message Builder
☒	3	name_family	Message Builder
☒	4	gender	Message Builder
☒	5	gender_code	Message Builder
☒	6	gender_display	Message Builder
☒	7	birthdate	Message Builder
☒	8	deceased	Message Builder
☒	9	name_family	Message Builder

Step \ Generated Script \

Message Segment:

Mapping:

Default Value:

String Replacement:

Regular Expression	Replace With

New Delete

b) Gender

The complete Transformer step for the attribute **gender** of the patient will then look as follows:

Edit Channel - MII-Patient Channel complete - Source Transformer

Enabled	#	Name	Type
☒	0	identifier_system	Message Builder
☒	1	identifier_value	Message Builder
☒	2	name_use	Message Builder
☒	3	name_family	Message Builder
☒	4	gender	Message Builder
☒	5	gender_code	Message Builder
☒	6	gender_display	Message Builder
☒	7	birthdate	Message Builder

Step \ Generated Script \

Message Segment:

Mapping:

Default Value:

String Replacement:

Regular Expression	Replace With

New Delete

1.5.2) MII Encounter

a) Record number

Here, the record number is in the FHIR message template identifier.value.

Edit Channel - MII-Encounter Channel complete - Source Transformer

Enabled	#	Name	Type
☒	0	aufnahmenummer	Message Builder
☒	1	class_code	Message Builder
☒	2	class_display	Message Builder
☒	3	serviceprovider	Message Builder
☒	4	period_start	Message Builder
☒	5	period_end	Message Builder
☒	6	admitsource	Message Builder
☒	7	dischargedisposition	Message Builder

Step \ Generated Script \

Message Segment:

Mapping:

Default Value:

String Replacement:

Regular Expression	Replace With

New Delete

b) Reference

Edit Channel - MII-Encounter Channel complete - Source Transformer

Enabled	#	Name	Type
☒	1	class_code	Message Builder
☒	2	class_display	Message Builder
☒	3	serviceprovider	Message Builder
☒	4	period_start	Message Builder
☒	5	period_end	Message Builder
☒	6	admitsource	Message Builder
☒	7	dischargedisposition	Message Builder
☒	8	reference	Message Builder

Step \ Generated Script \

Message Segment:

Mapping:

Default Value:

String Replacement:

Regular Expression	Replace With

New Delete

Since each case references a patient, each reference should begin with 'Patient?identifier='

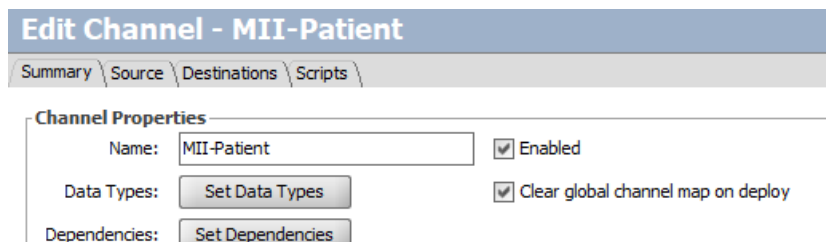
Important: Only the first part 'Patient?identifier=' is enclosed in single quotation marks ('..'). The rest is not!

Therefore, the mapping is: 'Patient?identifier=' + msg['row'][0]['Subject'].toString()

1.6) Deploying the channels and loading the data

Under "Channel Tasks," click "Redeploy All" to activate both channels.

Important: Both channels should have the status "Enabled." You can change this under:



Edit Channel - MII-Patient

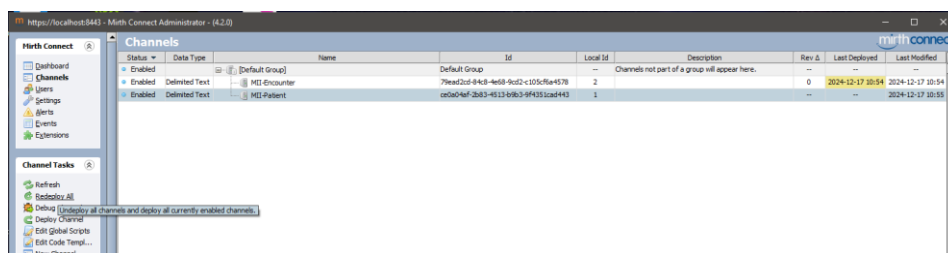
Summary | Source | Destinations | Scripts

Channel Properties

Name: ☒ Enabled

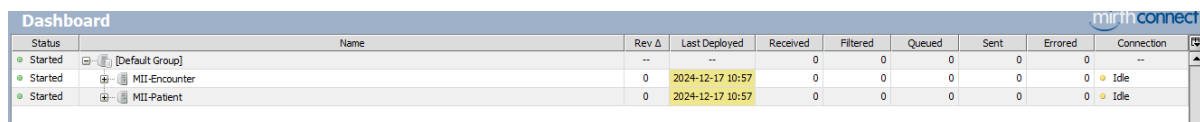
Data Types: ☒ Clear global channel map on deploy

Dependencies:



Status	Data Type	Name	Id	Local Id	Description	Rev Δ	Last Deployed	Last Modified
Enabled	[Default Group]	[Default Group]	Default Group	--	Channels not part of a group will appear here.	--	--	--
Enabled	Delimited Text	MII-Encounter	7f6ed2a2-8fcd-4e68-9e02-c103cfae6378	2		0	2024-12-17 10:54	2024-12-17 10:54
Enabled	Delimited Text	MII-Patient	ce88e7ef-28d3-4632-9b02-9f49322a2f43	1		0	2024-12-17 10:55	2024-12-17 10:55

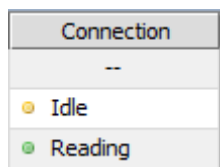
Deployed Channels:



Status	Name	Rev Δ	Last Deployed	Received	Filtered	Queued	Sent	Errored	Connection
Started	[Default Group]	--	--	0	0	0	0	0	--
Started	MII-Encounter	0	2024-12-17 10:57	0	0	0	0	0	Idle
Started	MII-Patient	0	2024-12-17 10:57	0	0	0	0	0	Idle

Once these are deployed, you can start copying the patient CSV file to the folder.

When Mirth-Connect reads the files, the following should be displayed:



Connection

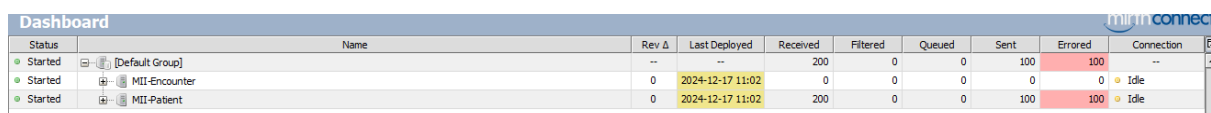
--

Idle

Reading

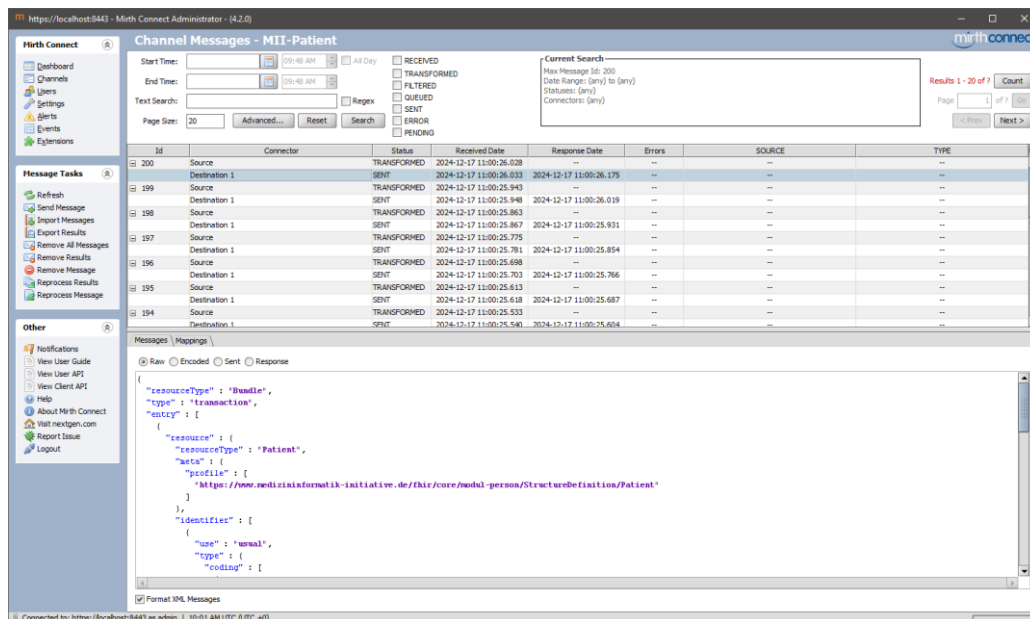
MII Patient:

100 patients should be sent successfully (Sent):



Status	Name	Rev Δ	Last Deployed	Received	Filtered	Queued	Sent	Errored	Connection
Started	[Default Group]	--	--	200	0	0	100	100	--
Started	MII-Encounter	0	2024-12-17 11:02	0	0	0	0	0	Idle
Started	MII-Patient	0	2024-12-17 11:02	200	0	0	100	100	Idle

Under "View Messages," you can view the sent messages and see any errors:



Test: To test whether all patients have been successfully loaded onto Hapi, enter http://localhost:8090/fhir/Patient?_summary=count in your browser:

HTTP 200 OK

Response Headers

```
X-Powered-By: HAPI FHIR 7.2.0 REST Server (FHIR Server; FHIR 4.0.1/R4)
X-Request-ID: sLNGzeSMLUxZNAti
```

Response Body

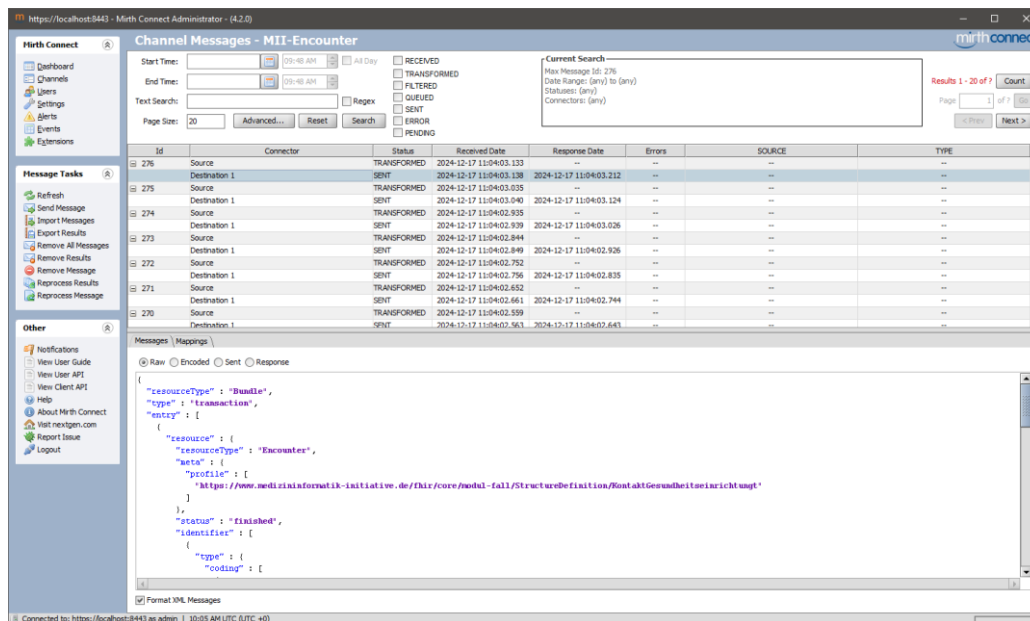
```
1  {
2    "resourceType": "Bundle",
3    "id": "99caa750-bb32-4baf-8746-bb1c84d758c6",
4    "meta": {
5      "lastUpdated": "2024-12-17T10:27:01.071+00:00",
6      "tag": [ {
7        "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationValue",
8        "code": "SUBSETTED",
9        "display": "Resource encoded in summary mode"
10     } ]
11  },
12  "type": "searchset",
13  "total": 100
14 }
```

MI1 Encounter:

275 cases should be sent successfully (Sent):

Dashboard										
Status	Name	Rev Δ	Last Deployed	Received	Filtered	Queued	Sent	Errored	Connection	
Started	[Default Group]	--	--	476	0	0	375	101	--	
Started	MI1-Encounter	0	2024-12-17 11:04	276	0	0	275	1	Idle	
Started	MI1-Patient	0	2024-12-17 11:04	200	0	0	100	100	Idle	

Under "View Messages," you can view the messages that were sent and see any errors:



Test: To test whether all patients have been successfully loaded onto Hapi, enter http://localhost:8090/fhir/Encounter?_summary=count in your browser.

2.1) Configuring ChlinFHIR

Change the data server at the top right under "Edit" and select "Public HAPI R4 Server" from the drop-down menu and confirm with "Save."

Current servers
Edit

Data Server	Public HAPI R4 server	?
Conformance Server	Public HAPI R4 server	?
Terminology Server	OntoserverR4 (terminology)	?

Add Server
Set all the same as the Data Server

Important: Before the local FHIR server can be added, you must change the data server from "InterSystems FHIR Server R4" to "Public HAPI R4 Server."

Then add the FHIR server under "Add Server":

URL: <http://localhost:8090/fhir/>

Add new server

Name

Uri

FHIR Version R4

Support \$everything ☒

Is Terminology Server ☐

4.0.1

The name of the server (will be displayed in server selection dialogs)

The full uri to the server (including port). Include a trailing /

The version of FHIR that the server supports

Set true if this server supports the \$everything operation against Patient (Needed for Patient Data viewer)

Set true if this server supports Terminology services (especially \$expand)

Cancel
Add

Then select the newly added FHIR server from the list of all data servers under "Edit Server" and confirm with "Save."

Current servers		Edit
Data Server	Mi-Lab Hapi	?
Conformance Server	Mi-Lab Hapi	?
Terminology Server	OntoserverR4 (terminology)	?
Add Server		Set all the same as the Data Server

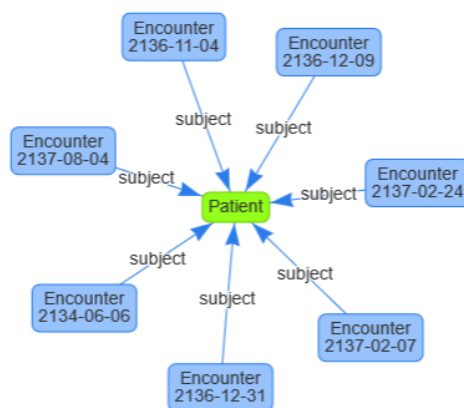
2.2) ChlinFHIR Patient Viewer

2.2.1) How old is the patient with ID 58 (identifier: 10015860)?

Solution: 53 years old (date of birth: January 1, 1971)

2.2.2) How many cases are there for the patient with ID 2 (identifier: 10003400)?

Solution: Resource Reference Graph



Solution: 7 cases

2.2.3) Is the patient with ID 2 (identifier: 10003400) already deceased?

Solution: Yes (deceasedBoolean: false)

2.3) ChlinFHIR Server Query

2.3.1) How many patients are female?

Solution: *Patient?gender=female&_count=100*

http://localhost:8090/fhir/Patient?gender=female&_count=100

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2.3.2) How many patients have already died?

Solution: *Patient?deceased=true&_count=100*

```
http://localhost:8090/fhir/Patient?deceased=true&_count=100
```

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2.3.3) How many cases belong to patients who have already died?

Solution: Encounter?patient.deceased= true&_count=200

```
http://localhost:8090/fhir/Encounter?patient.deceased=true&_count=200
```

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